

Camelot Progress Presentation

Software Engineering

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March 24th, 2017

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About the Camelot Server

- Camelot is an asynchronous *Python3* server.
- The server is housed on a Raspberry Pi (as provided by Ian Howell) so that the client team will be able to interact and test their code on it (assuming the server is up and running) and also so that interaction with the server can be done remotely.
- The database is being tackled specifically by a module in Python called *psycpg2* which uses *PostgreSQL*.
- All communication between the server and client applications is done via *JSON* encoded data.

Camelot Inspiration



Figure 1: “On second thought, let’s not go to Camelot. ’Tis a silly place.” — Arthur, Monty Python

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- Changes** Sometimes when you innovate, you make mistakes (i.e. choices for users. . .). It is best to admit them quickly, and get on with improving your other innovations.

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Tools By picking some of the best tools (Github, $\text{\LaTeX}$, PostgreSQL), it has made for a much better workflow.

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- Server code is written in Python3.
 - Documentation is written in Doxygen.
- All development is on Github.
 - This includes meeting notes, documentation, and all production code (i.e. database and server).
 - Github issues/milestones for task management.
 - Contributions to make sure even workload.
- All documentation/presentation is written in $\text{\LaTeX}$.
- Using PostgreSQL for the database.
- Using Discord for team/client chat – by any memes necessary.

Documentation Demo

Python enables easy server setup.

```
1  def main():
2      server = ThreadedTCPServer(("0.0.0.0", 9009), ThreadedTCPRequestHandler)
3      ip, port = server.server_address
4      server.socket.listen(10)
5
6      server_thread = threading.Thread(target=server.serve_forever)
7      server_thread.dameon = True
8      server_thread.start()
9      print("Server loop running in thread:", server_thread.name)
10
11     try:
12         # Loop forever
13         while True:
14             pass
15     except KeyboardInterrupt:
16         print("Cleaning up server..")
17         server.shutdown()
18         server.server_close()
19         print("Done! Goodbye")
20
21
```

Server Demo

Github Demo

Some of the challenges we encountered this far are:

1. Communication between the client and server teams.

Description & Solution

Even though we have set up a system for communicating between the client and server teams (Discord), there is little to no talk between the teams. *Solution: Communicate more..?*

Some of the challenges we encountered this far are:

1. Communication between the client and server teams.
2. Direction and vision of the product.

Description & Solution

Although there is an established, core vision (IRC clone), working out the details has proven to be difficult. *Solution: Break it down to basic principles and work with teammates.*

Challenges

Some of the challenges we encountered this far are:

1. Communication between the client and server teams.
2. Direction and vision of the product.
3. Learning curve of tools, teammates, and project.

Description & Solution

Because we're relatively new to each other, and are unaware of each other's skill sets, assigning tasks becomes tricky. Also learning how to implement new things can be difficult. *Always be sure teammates are comfortable with the tasks they're given.*

- Encryption.
 - Still debating if we'll be using ROT13 (znlor vs jr tvq thq, jr'yy hfr EBG26.) or an in-house, post-quantum cryptographic hash function using a multivariate-quadratic public-key signature system.
- Open source

- Encryption.
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- Open source at the end of the semester.
- Full documentation guide courtesy of Doxygen.

Current and Future Plans

1. Consolidate the client and server teams to get unified protocol and systems specification document.
2. Finish coding the actual chatroom.
 - Server
 - Database
 - Script to automate a chatroom environment
3. Finish documenting the chatroom (possibly API guide).

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4. ...
5. Profit

In Closing

Spam Illya with all questions, comments, and insults.

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Special thanks to our awesome team.

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 Hunter Mathews

 Illya Starikov

 William Thurman

 Zachary Wileman